

Université d'Ottawa · University of Ottawa

Faculté des sciences
Mathématiques et de statistique

Faculty of Science
Mathematics and Statistics

MAT 1302A: Mathematical Methods II
Professor : Aaron Tikuisis

Final Exam - Winter 2019
April 24, 2019, 9:30–12:30

Surname _____ First Name _____

Student # _____ Seat # _____

Instructions: (Please read them attentively)

- Write your full name, student number and assigned seat number in the designated space above.
- The duration of this exam is **180 minutes**.
- This is a closed book exam that contains **17 questions**. The **first 12** questions are multiple choice, and the **last 5** are long answers.
- For questions 1 to 12, you must **write** your answers in the table on the second page.
- For questions 13 to 17, you must clearly justify your answers.
- No calculators are permitted.
- There are two additional blank pages at the end of this exam that you may use as scrap paper.
- Cellular phones, unauthorized electronic devices or course notes are not allowed during this exam. Phones and devices must be turned off and put away in your bag. Do not keep them in your possession, such as in your pockets. If caught with such a device or document, the following may occur: academic fraud allegations will be filed which may result in your obtaining a **0** (zero) for the exam.

Good luck!!!

By signing below, you acknowledge that you have read and ensured that you are complying with the above statements.

Signature: _____

Write your answers to the multiple choice questions in the following table.

Question	1	2	3	4	5	6	7	8	9	10	11	12
Answer												
Note												

Do not write in the following table.

Question	1–12	13	14	15	16	17	Total
Maximum	24	7	5	7	8	6	57
Note							

Part A: multiple choice questions

For questions 1–12, **write your answers in the table on page 2**. *Each question is worth* **2 points**.

1. Let $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 2 & -1 \end{bmatrix}$. The second row of the matrix $A^T B$ is

(A) $\begin{bmatrix} -2 & -1 \end{bmatrix}$
(B) $\begin{bmatrix} 2 & 2 & 4 \end{bmatrix}$
(C) $\begin{bmatrix} 1 & 5 & -1 \end{bmatrix}$
(D) $\begin{bmatrix} -2 & 4 & 2 \end{bmatrix}$
(E) $\begin{bmatrix} -2 & 1 & -5 \end{bmatrix}$
(F) The operation is not defined.

Solution: (C) because $A^T B = \begin{bmatrix} 2 & 3 & 0 \\ 1 & 5 & -1 \end{bmatrix}$

2. Consider the complex number $z = \dfrac{1-i}{1+i}$. Its expression in the form $a + bi$ is given by

(A) $z = 1 - i$
(B) $z = 1 + i$
(C) $z = -\dfrac{1}{2} - i\dfrac{1}{2}$
(D) $z = 0 - i$
(E) $z = 1 - 2i$
(F) $z = 0 + i$

Solution: (D) $z = -i$.

3. Let $A = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 1 & 0 & 1 \end{bmatrix}$. The third row of A^{-1} is

(A) $\begin{bmatrix} -\frac{1}{2} & 0 & \frac{1}{2} \end{bmatrix}$.

(B) $[-1 \;\; 0 \;\; 1]$.

(C) $[0 \;\; 0 \;\; -1]$.

(D) $[1 \;\; 0 \;\; 1]$.

(E) $\begin{bmatrix} \frac{1}{2} & 1 & \frac{1}{2} \end{bmatrix}$.

(F) A is not invertible.

Solution: (A)

4. Let $A = \begin{bmatrix} 1 & 5 & 1 & 4 & 5 \\ 0 & 1 & 3 & 2 & 6 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$. The dimensions of $\mathcal{C}(A)$ and of $\mathcal{N}(A)$ are given by

(A) $\dim(\mathcal{C}(A)) = 2$ and $\dim(\mathcal{N}(A)) = 3$.
(B) $\dim(\mathcal{C}(A)) = 3$ and $\dim(\mathcal{N}(A)) = 2$.
(C) $\dim(\mathcal{C}(A)) = 3$ and $\dim(\mathcal{N}(A)) = 1$.
(D) $\dim(\mathcal{C}(A)) = 4$ and $\dim(\mathcal{N}(A)) = 1$.
(E) $\dim(\mathcal{C}(A)) = 5$ and $\dim(\mathcal{N}(A)) = 0$.
(F) $\dim(\mathcal{C}(A)) = 5$ and $\dim(\mathcal{N}(A)) = 4$.

Solution: (B) $\dim(\mathcal{C}(A)) = 3$ et $\dim(\mathcal{N}(A)) = 2$.

5. Let A be a 3×3 matrix. Assume the characteristic polynomial of A is given by $p_A(x) = -x^3 + x$. Which one of the following statements is true?

(A) A is invertible.
(B) $\det A = 1$.
(C) $\lambda = 2$ is an eigenvalue of A^2.
(D) $\lambda = 3$ is an eigenvalue of A.
(E) The eigenvalues of A^4 are 1 and 0.
(F) The eigenvalues of A^4 are -1 and 2.

Solution: (E)

6. Let $A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$, $B = \begin{bmatrix} a & b & c \\ g & h & i \\ 2d-a & 2e-b & 2f-c \end{bmatrix}$.
Find $\det(B)$ knowing that $\det(A) = 5$.

(A) $\det(B) = -5$.
(B) $\det(B) = 5$.
(C) $\det(B) = 10$.
(D) $\det(B) = -10$.
(E) $\det(B) = -2$.
(F) $\det(B) = 2$.

Solution: (D) $\det(B) = (-2) \times 5 = -10$.

7. Let A be a 5×4 matrix and $\mathbf{b}$ be a vector of $\mathbb{R}^5$. Assume the matrix equation $A\mathbf{x} = \mathbf{b}$ has infinitely many solutions. Which one of the following statements **is not true**?

(A) The vector $\mathbf{b}$ is a linear combination of the columns of A.
(B) Each row of the reduced row echelon form of A contains a leading one.
(C) The columns of A are linearly dependent.
(D) $\operatorname{rank} A < 4$.
(E) $\mathcal{N}(A) \neq \{\mathbf{0}\}$.
(F) $\dim(\mathcal{N}(A)) > 0$.

Solution: (B)

8. Which one of the following subsets is a subspace of $\mathbb{R}^3$?

(A) $\left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \;\middle|\; x_1 + x_2 - x_3 = 1 \right\}$

(B) $\left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \;\middle|\; xy = z \right\}$

(C) $\left\{ \begin{bmatrix} a+b \\ b \\ 2b \end{bmatrix} \;\middle|\; a, b \in \mathbb{R} \right\}$

(D) $\left\{ \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \right\}$

(E) $\left\{ \begin{bmatrix} a+b \\ b+1 \\ a-b \end{bmatrix} \;\middle|\; a, b \in \mathbb{R} \right\}$

(F) $\left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \;\middle|\; \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}$

Solution: (C).

9. Find the values of h and k for which the linear system

$$\begin{cases} 2x_1 + hx_2 = -2 \\ x_1 + 3x_2 = k \end{cases}$$

has infinitely many solutions.

(A) $h = 0$ and $k = -3$.
(B) $h = 1$ and $k = 3$.
(C) $h = 1$ and $k = -6$.
(D) $h = 6$ and $k = -1$.
(E) $h = 3$ and $k = 1$.
(F) $h = 3$ and $k = 0$.

Solution: (D)

10. Let A be an $n \times n$ matrix. Which one of the following statements is **true** for A if $\mathcal{N}(A) = \{\mathbf{0}\}$?

(A) $\det(A) = 0$.
(B) The linear system $A\mathbf{x} = \mathbf{0}$ has infinitely many solutions.
(C) The columns of A are linearly dependent.
(D) The reduced row echelon form of A has $n - 1$ pivot columns.
(E) A^T is invertible.
(F) $\mathcal{C}(A) = \{\mathbf{0}\}$.

Solution: (E).

11. An economy is divided into two sectors: Energy and Services. Suppose the following:
- To produce one unit of output, the Energy sector consumes 0.8 units from its own sector and 0.3 units of Services.
- For each unit of output, Services requires 0.6 units from its own sector and 0.2 units of Energy.

What are the production levels required to meet a final demand of 20 units from the Energy sector and 30 units from the Services sector?

(A) 350 units from Energy and 250 units from Services.
(B) 700 units from Energy and 600 units from Services.
(C) 600 units from Energy and 700 units from Services.
(D) 550 units from Energy and 300 units from Services.
(E) 0 units from Energy and 700 units from Services.
(F) 700 units from Energy and 0 units from Services

Solution: (B). $C = \begin{bmatrix} .8 & .2 \\ .3 & .6 \end{bmatrix}$ so $I - C = \begin{bmatrix} .2 & -.2 \\ -.3 & .4 \end{bmatrix} \Rightarrow (I-C)^{-1} = \frac{1}{.02}\begin{bmatrix} .4 & .2 \\ .3 & .2 \end{bmatrix} = \begin{bmatrix} 20 & 10 \\ 15 & 10 \end{bmatrix}$

$\Rightarrow \mathbf{x} = \begin{bmatrix} 20 & 10 \\ 15 & 10 \end{bmatrix}\begin{bmatrix} 20 \\ 30 \end{bmatrix} = \begin{bmatrix} 700 \\ 600 \end{bmatrix}$

and production levels are $\begin{cases} \text{Energy:} & 700 \\ \text{Services:} & 600. \end{cases}$

12. Let $A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$. The eigenvalues of A are

(A) $-i$ and i
(B) $1-i$ and $1+i$
(C) -1 and 1
(D) 1 and i
(E) 0 and $1-i$
(F) $2i$ and $-2i$

Solution: We have

$$p_A(x) = \det(A - xI) = x^2 + 1 = (x-i)(x+i).$$

So the eigenvalues are $\pm i$.

Part B: Long Answer Questions

For questions 13–17, you must clearly justify your answers.

13. **(7 points)** Determine whether the following linear system is consistent. **If it is**, find its general solution and express it in **vector parametric form**.

$$\begin{cases} x_1 + 3x_2 + 9x_3 - 2x_4 = 3 \\ -2x_1 - 6x_2 - 18x_3 + 4x_4 = -6 \\ 2x_1 + 7x_2 + 23x_3 = 2 \end{cases}$$

Solution: We row reduce the augmented matrix. We have

$$[A|b] = \left[\begin{array}{cccc|c} 1 & 3 & 9 & -2 & 3 \\ -2 & -6 & -18 & 4 & -6 \\ 2 & 7 & 23 & 0 & 2 \end{array}\right] \xrightarrow{\begin{matrix} R_2 + 2R_1 \to R_2 \\ R_3 - 2R_1 \to R_3 \end{matrix}} \left[\begin{array}{cccc|c} 1 & 3 & 9 & -2 & 3 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 5 & 4 & -4 \end{array}\right]$$

$$\xrightarrow{R_2 \leftrightarrow R_3} \left[\begin{array}{cccc|c} 1 & 3 & 9 & -2 & 3 \\ 0 & 1 & 5 & 4 & -4 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{R_1 - 3R_2 \to R_1} \left[\begin{array}{cccc|c} 1 & 0 & -6 & -14 & 15 \\ 0 & 1 & 5 & 4 & -4 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right].$$

The system is consistent, and the general solution is:

$$\Rightarrow \begin{cases} x_1 = 6x_3 + 14x_4 + 15 \\ x_2 = -5x_3 - 4x_4 - 4 \\ x_3 : \text{free} \\ x_4 : \text{free} \end{cases}.$$ The vector parametric form is

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 6x_3 + 14x_4 + 15 \\ -5x_3 - 4x_4 - 4 \\ x_3 \\ x_4 \end{bmatrix} = x_3 \begin{bmatrix} 6 \\ -5 \\ 1 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 14 \\ -4 \\ 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 15 \\ -4 \\ 0 \\ 0 \end{bmatrix}, x_3, x_4 \in \mathbb{R}.$$

14. **(5 points)** For which values of k is the vector $\mathbf{b} = \begin{bmatrix} 1 \\ k \\ 1 \end{bmatrix}$ a linear combination of $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $\begin{bmatrix} 4 \\ 2 \\ 6 \end{bmatrix}$, $\begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$?

Solution:

$$\left[\begin{array}{ccc|c} 1 & 4 & 1 & 1 \\ 2 & 2 & 1 & k \\ 3 & 6 & 2 & 1 \end{array}\right] \xrightarrow[R_3 \to R_3 - 3R_1]{R_2 \to R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 4 & 1 & 1 \\ 0 & -6 & -1 & k-2 \\ 0 & -6 & -1 & -2 \end{array}\right]$$

$$\xrightarrow{R_3 \to R_3 - R_2} \left[\begin{array}{ccc|c} 1 & 4 & 1 & 1 \\ 0 & -6 & -1 & k-2 \\ 0 & 0 & 0 & -k \end{array}\right] \Rightarrow \; k = 0.$$

15. The market of a cosmetic product is divided into two competitors, **Alpha** and **Beta**. Suppose that Beta has just entered the market and has 10% of the market shares, which means Alpha has the remaining 90%. Analysts predict that every year, 20% of Alpha clients will switch to Beta, and that 10% of Beta clients will switch to Alpha.

(a) **(1 point)** Write down the migration matrix P.

Solution: $P = \begin{bmatrix} .8 & .1 \\ .2 & .9 \end{bmatrix}$ (Note that another possibility is $\begin{bmatrix} .9 & .2 \\ .1 & .8 \end{bmatrix}$

(b) **(1 point)** Determine the market shares of Alpha and Beta after one year according to the analysts' predictions.

Solution: $\mathbf{x}_1 = P\mathbf{x}_0 = \begin{bmatrix} .8 & .1 \\ .2 & .9 \end{bmatrix} \begin{bmatrix} 0.9 \\ 0.1 \end{bmatrix} = \begin{bmatrix} .73 \\ .27 \end{bmatrix}$

Therefore after 1 year the market shares are $\begin{cases} \text{Alpha:} & 73\% \\ \text{Beta:} & 27\% \end{cases}$

(c) **(5 points)** Suppose that Alpha and Beta don't change their marketing strategies and that the migration between the two competitors remains the same. What will the market shares be in the long run?

Solution:

$P\mathbf{x} = \mathbf{x} \;\Rightarrow\; (P - I)\mathbf{x} = \mathbf{0} \;\Rightarrow\; \begin{bmatrix} -.2 & .1 \\ .2 & -.1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$$\left[\begin{array}{cc|c} -.2 & .1 & 0 \\ .2 & -.1 & 0 \end{array}\right] \xrightarrow{R_2 \to R_2 + R_1} \left[\begin{array}{cc|c} -.2 & .1 & 0 \\ 0 & 0 & 0 \end{array}\right] \xrightarrow{R_1 \to -\frac{10}{2}} \left[\begin{array}{cc|c} 1 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 \end{array}\right] \Rightarrow \begin{cases} x_1 = \frac{1}{2}x_2 \\ x_1 + x_2 = 1 \end{cases}$$

$\Rightarrow\; x_2 = \frac{2}{3} \;\Rightarrow\; x_1 = \frac{1}{3}$

Therefore the long term market shares are: $\begin{cases} \text{Alpha: } (100/3)\% \\ \text{Beta: } (200/3)\% \end{cases}$

16. Let $A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 3 & 0 \\ 1 & 0 & 2 \end{bmatrix}$.

(a) **(2 points)** Find the characteristic polynomial $p_A(x)$ of A and **use this to deduce** that the eigenvalues of A are 1 and 3. *You must justify all details of your answer to receive full marks.*

Solution:
By expanding along the second column we have:

$$\begin{aligned} \det(A-\lambda I) &= \begin{vmatrix} 2-\lambda & 0 & 1 \\ 0 & 3-\lambda & 0 \\ 1 & 0 & 2-\lambda \end{vmatrix} = (3-\lambda)\begin{vmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{vmatrix} \\ &= (3-\lambda)\left((2-\lambda)(2-\lambda)-1\right) = (3-\lambda)(\lambda^2-4\lambda+3) \\ &= (3-\lambda)(\lambda-1)(\lambda-3) = (3-\lambda)^2(1-\lambda) \end{aligned}$$

and therefore the eigenvalues of A are $\lambda = 1$, and $\lambda = 3$.

(b) **(2 points)** Find a basis for the eigenspace associated to the eigenvalue $\lambda = 1$ of A.

Solution:
For $\mathcal{E}_1$ we have

$$A - 1I = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow \quad v_1 = -v_3 \text{ and } v_2 = 0,\ v_3 \text{ free}$$

$$\Rightarrow \quad \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} -v_3 \\ 0 \\ v_3 \end{bmatrix} = v_3 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

$$\Rightarrow \text{ basis for the eigenspace } \mathcal{E}_1 \text{ of } \lambda = 1 \ : \ \left\{ \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

See the next page for the rest of this question.

(c) **(2 points)** Find a basis for the eigenspace associated to the eigenvalue $\lambda = 3$ of A.

Solution: For $\mathcal{E}_3$ we have

$$A - 3I = \begin{bmatrix} -1 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow \quad v_1 = v_3\,, v_2 \text{ and } v_3 \text{ free}$$

$$\Rightarrow \quad \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 0 \\ v_2 \\ 0 \end{bmatrix} + \begin{bmatrix} v_3 \\ 0 \\ v_3 \end{bmatrix} = v_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + v_3 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

$$\Rightarrow \text{ basis for the eigenspace } \mathcal{E}_3 \; of \lambda = 3 \; : \; \left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

(d) **(2 points)** Is the matrix A diagonalizable? If so, write down an invertible matrix P and a diagonal matrix D such that $P^{-1}AP = D$. Justify your answer.

Solution: Yes, the matrix is diagonalizable because $\dim \mathcal{E}_1 + \dim \mathcal{E}_3 = 3$ multiplicity of λ for either eigenvlues 1 and 3.

Let

$$P = \begin{bmatrix} -1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \Rightarrow D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}.$$

17. Consider the following subset of $\mathbb{R}^3$

$$H = \text{span}\left\{ \begin{bmatrix} -2 \\ 1 \\ -3 \end{bmatrix}, \begin{bmatrix} 4 \\ -2 \\ 6 \end{bmatrix}, \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix} \right\}.$$

(a) **(1 point)** Explain in one sentence why H is a subspace of $\mathbb{R}^3$.

(b) **(5 points)** Find a basis of H and determine $\dim(H)$.

Solution: (a): Since H is a span of a set of vectors, it is a subspace.
(b): Let

$$A = \begin{bmatrix} -2 & 4 & 3 & 1 \\ 1 & -2 & 0 & 1 \\ -3 & 6 & 1 & -2 \end{bmatrix},$$

so that $A = \mathcal{C}(A)$.
Row reduction:

$$A = \begin{bmatrix} -2 & 4 & 3 & 1 \\ 1 & -2 & 0 & 1 \\ -3 & 6 & 1 & -2 \end{bmatrix} \xrightarrow[3R_2 + R_3 \to R_3]{\substack{R_2 \to R_1 \\ R_1 + 2R_2 \to R_2}} \begin{bmatrix} 1 & -2 & 0 & 1 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & 1 & 1 \end{bmatrix} \xrightarrow{R_3 - 3R_2 \to R_3} \begin{bmatrix} 1 & -2 & 0 & 1 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

From this we see that the first and third columns are the pivot columns, so these form our basis, which is

$$\left\{ \begin{bmatrix} -2 \\ 1 \\ -3 \end{bmatrix}, \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

Hence $\dim(H) = 2$.

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Extra page to be used as scrap paper. 2